

PROFESSIONAL PROFILE

Ph.D. student in Electrical Engineering at Sharif University of Technology, with an M.Sc. in Electrical Engineering, specializing in Digital Electronic Systems from Shahid Beheshti University. Skilled in FPGA/ASIC design, embedded systems, and hardware/software co-design, with hands-on experience in VHDL, HLS, and system-level integration using Xilinx tools, HSPICE, and Cadence Virtuoso. Master's thesis focused on designing approximate computing circuits to enhance spiking neural networks, laying a solid foundation for research in neuromorphic and energy-efficient AI hardware. Equipped with a solid academic background in digital architecture, low-power circuit design, and intelligent computing systems, and highly motivated to contribute to advanced research in next-generation hardware for artificial intelligence.

RESEARCH INTERESTS

- Neuromorphic computing
- Approximate computing
- Artificial Intelligence (AI)
- Spiking Neural Networks (SNNs)
- Internet of Things (IoT)
- Digital Circuit Design

EDUCATION

Ph.D. in Electrical Engineering | October 2025 – Present
Sharif University of Technology, Tehran, Iran

M.Sc. in Electrical Engineering, Specialization in Digital Electronic Systems | October 2022 – February 2025
Shahid Beheshti University, Tehran, Iran
Thesis: Designing Approximate Computing Circuits for Developing Spiking Neural Networks
Supervisor: Dr. Somayeh Timarchi
Cumulative GPA: 17.99 out of 20 (89.95%)

B.Sc. in Electrical Engineering | October 2018 – September 2022
Shahid Beheshti University, Tehran, Iran
Thesis: Design and implementation of a complete smart IoT system for monitoring and controlling laboratory events
Supervisor: Dr. Hadi Asharioun
Cumulative GPA: 17.84 out of 20 (89.2%)
Rank: 3rd in program

PUBLICATIONS

- **R. Badiei**, S. Timarchi and A. Zakaleh, "Low-Power Resource-Efficient FPGA Implementation of Modified FitzHugh–Nagumo Neuron for Spiking Neural Networks," *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 72, no. 11, pp. 1780-1784, Nov. 2025, doi: 10.1109/TCSII.2025.3615935.
- A. Hajjafari, **R. Badiei**, H. Asharioun and R. Nasiri, "Design and Implementation of an IoT-based Smart Home with the Ability to Communicate with the Smart Grid," in *2023 International Conference on Protection and Automation of Power Systems (IPAPS)*, Tehran, Iran, Islamic Republic of, 2023, pp. 1-5, doi: 10.1109/IPAPS58344.2023.10123324.
- A. Zakaleh, **R. Badiei**, Z. H. Zadeh and S. Timarchi, "Low-Power Hyperbolic CORDIC Design of the Fitzhugh-Nagumo Neuron for Neuromorphic Systems," *IEEE Embedded Systems Letters*, accepted for publication, 2026, doi: 10.1109/LES.2026.3710888.

ACADEMIC SERVICE

- Reviewer for Neurocomputing (Elsevier)

EXPERIENCE

Research

2023 – 2026 | Research Assistant — Shahid Beheshti University, Tehran, Iran

Conducted research on neuromorphic computing and Spiking Neural Networks (SNNs) under the supervision of Professor Somayeh Timarchi, focusing on energy- and resource-efficient hardware implementations.

- Designed and simulated SNNs for classification with STDP learning.
- Developed modified versions of Izhikevich, FitzHugh-Nagumo, and Hindmarsh-Rose models to improve energy and resource efficiency.
- Conducted experiments to optimize energy and resource usage in neuromorphic hardware.
- Used VHDL for circuit modeling and Python for simulation and analysis.
- Contributed to research papers on neuromorphic computing.

2022 | Electrical Engineer (IoT Designer & Implementer), Research Assistant — Shahid Beheshti University, Tehran, Iran

Developed a smart IoT system for monitoring and controlling lab events as part of my bachelor's project. Integrated diverse devices through a custom communication network.

- Developed communication protocols for seamless device integration.
- Built a real-time IoT dashboard accessible locally and globally.

Teaching

2022 | Teaching Assistant, Electrical Engineering Course — Shahid Beheshti University, Tehran, Iran

Served as a Teaching Assistant for the C++ programming course under Professor Hadi Asharioun. Developed instructional materials and provided support to students to enhance their programming skills.

- Assisted in explaining complex programming concepts and fostering collaborative learning.
- Guided students in C and C++ coding, emphasizing logic and flowchart integration.
- Helped students master syntax and practical coding applications.

Work

May 2026 – Present | Electronics Engineer — ASHA Beam Profile, Tehran, Iran

Worked on embedded electronic systems involving hardware design, microcontrollers, Embedded Linux, camera integration, and FPGA-based development.

- Designed custom PCBs for embedded systems, including a daughterboard for the Cubie A7S platform featuring an STM32G031 microcontroller.
- Developed embedded systems and firmware using STM32 and ESP32 microcontrollers, including hardware/software integration and debugging.
- Worked with Raspberry Pi 3 and Cubie A7S platforms; adapted existing Linux drivers and configured V4L2 and Device Tree for MIPI CSI-2 camera integration. Developed a Qt-based application for camera register configuration, image acquisition, processing, and visualization.
- Worked on FPGA-based system development, hardware implementation, and hardware/software integration.

Aug 2025 – Apr 2026 | Medical Equipment Supplier — Behbud Darman, Tehran, Iran

Worked in the supply and sale of medical equipment, including customer communication, product presentation, and sales activities.

2021 – 2023 | IoT Designer & Implementer — Smart Home Venture, Tehran, Iran

Developed and deployed an IoT-based smart-home prototype as an entrepreneurial venture, working across embedded hardware, wireless communication, and full-stack software.

- Designed and implemented embedded systems for smart devices, including sensor interfacing, PCB layout, and microcontroller programming.
- Developed a full-stack web application for real-time control and monitoring of smart devices using React.js and Node.js.
- Integrated hardware and software components using MQTT and TCP/IP for remote monitoring and automation.
- Collaborated with a small engineering team throughout the prototyping and deployment phases.

Jun 2021 – Nov 2021 | Electrical Engineer Intern — Andishe Negar Pars Company, Tehran, Iran

Collaborated in the development of smart home devices, including PCB design and programming for smart switches, temperature controllers, power outlets, and sensors (motion, gas, and flame detectors). Worked with the company's smart home system protocols to integrate devices seamlessly.

RELEVANT COURSEWORK

• VHDL	19.12/20	• Interface Circuits	19.5/20	• Linear ICs Design	19.5/20
• ASIC/FPGA Circuits	20/20	• Internet of Things	20/20	• Electronics II	20/20
• Advanced Digital Electronics	18.25/20	• Machine Learning	19.5/20	• Electronics III	19.9/20

SKILLS

Digital Design (FPGA/ASIC)

- Hardware Description Languages: VHDL
- High-Level Synthesis (HLS): Vivado HLS
- FPGA Tools: Xilinx ISE, Vivado Design Suite, ModelSim
- SoC: System-level integration, hardware/software co-design, IP integration
- ASIC Tools: HSPICE; Cadence Virtuoso

Artificial Intelligence

- Programming & Tools: Python, MATLAB
- Frameworks: TensorFlow, PyTorch, Scikit-Learn
- Concepts: Neural Networks (NNs), Machine Learning (ML), Deep Learning (DL)

Embedded Systems

- Programming Languages & IDEs: C, C++, Assembly, MicroPython; CodeVision IDE, Arduino IDE
- Microcontrollers & Platforms: STM8, STM32, ESP32, ESP8266, Arduino, ATmega
- Platforms: Raspberry Pi (Raspbian)
- RTOS & Real-Time Systems: FreeRTOS; real-time sensor/actuator interfacing
- Hardware & Circuit Design: PCB design & layout (Altium Designer); Circuit Design, Soldering, Wiring, System Integration
- Simulation Tools: Multisim, Proteus
- Communication Protocols (IP, TCP, MQTT, LAN, LoRaWAN, Mesh)
- Wireless Communication: RF, LoRa (RFM69/95), NRF24, Wi-Fi, Bluetooth, ZigBee
- Serial Protocols: UART, SPI, I2C

Full-stack Web Development

- Frontend: HTML, CSS, JavaScript, React.js, TypeScript, Next.js
- Backend: Node.js, Express.js, SQL
- DevOps & Deployment: Nginx, Linux shell scripting

SELECTED PROJECTS

- Designed and simulated a Human Activity Recognition system in a smart home environment using deep learning models such as CNN and LSTM in “Deep Learning course”. [June 2023]
- Simulated and implemented the paper "Hardware/Software Co-Design of a Dynamically Configurable SHA-3 System-on-Chip (SoC)" in “ASIC/FPGA Circuits course”. [June 2023]
- Designed and implemented a Direct Digital Synthesis (DDS) interface for the AD9951 IC using VHDL in “VHDL course”. [February 2023]
- Designed and implemented a voice recognition system using Gaussian Mixture Models (GMM) for smart access control in “Machine Learning course”. [June 2022]

ENGLISH LANGUAGE PROFICIENCY

- **Level:** Upper-Intermediate
- **IELTS Academic:** Overall Band 6.5 (August 2025)